Xinyi Yang

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EDUCATION

Columbia University Sept.2026 – Dec.2027(Expected)
M.S. in Artificial Intelligence; AI and Advanced Computing Concentration New York City, USA

University of California, Santa Barbara Sept.2025 – Jun.2026
Exchange Student; Electrical Computer Engineering Santa Barbara, USA

Huazhong University of Science and Technology Sept.2022 – Jun.2026
B.E. in Artificial Intelligence and Automation; Honors Class Wuhan, China

RESEARCH & PROFESSIONAL EXPERIENCE

Intelligent Perception and Auto-Driving Lab, HUST Jan. 2024 – Nov. 2025
Research Assistant (Undergraduate Researcher) Wuhan, China

- Developed a multimodal LLM agent by extending Monkey/Qwen-VL with external tool-use capabilities for perception, planning, and decision-making.
- Built a GPT-4-based automated instruction-data generation pipeline to synthesize high-quality tool-calling conversations for instruction tuning.
- Fine-tuned Qwen-VL with LoRA to enable external tool invocation, integrating Grounding DINO, SAM, and InstructPix2Pix into an end-to-end multimodal agent.
- Contributed to the development and evaluation of a tool-augmented multimodal LLM that achieved 45.4 on MM-VET, outperforming the LLaVA-Plus baseline by 29.7%.

CamAI+ Deep Learning and Neural Networks Program Jan. 2023 – Mar. 2023
Project Team Lead Cambridge, Online

- Developed a YOLOv5-based text-recognition pipeline; built a custom image-text dataset and evaluated performance under complex visual conditions.

China Telecom Corporation Limited Wuhan Branch Jun. 2025 – Sept. 2025 | Jul. 2026 – Sept. 2026
AI Algorithm Intern Wuhan, China

- **Financial Document Intelligence (Summer 2025).** Developed an intelligent document auditing system integrating sequence-to-sequence text generation with document understanding for automated financial document analysis.
- Redesigned the generation pipeline, reducing repetitive outputs from 27% to 9%, and fine-tuned Faster R-CNN for document understanding, achieving 0.78 mAP.
- **AI-based Examination Monitoring (Summer 2026).** Developed a real-time examination monitoring system using YOLO11 for abnormal behavior detection in surveillance videos.
- Optimized the detection pipeline with PyTorch to support efficient real-time inference in practical deployment scenarios.

PROJECTS

Simulation of Three-phase SPWM Inverter Power Supply Circuit June, 2025

- Modeled and simulated a three-phase SPWM inverter to analyze the impact of modulation and frequency parameters on output performance.
- Designed SPWM signal generation and conducted a waveform analysis by using MATLAB/Simulink.
- Analyzed the relationship between DC bus voltage, modulation depth, and output amplitude, correlating simulation data with analytical predictions.

Speed Setting and Design of PI Regulator June 2025

- Designed and implemented a Field-Oriented Control (FOC) system for a PMSM on an STM32G4 platform, integrating simulation and hardware testing.
- Developed speed reference generators, tuned PI regulators, and validated closed-loop performance through Simulink and real-time experiments.
- Achieved approximately 1.5% steady-state speed accuracy and improved my skills in motor control, embedded systems, and real-time simulation.

Design of PID Control System for Liquid Level of Single-Volume Water Tank April 2025

- Designed and implemented a PID control system for a single-tank process by using Siemens S7-1200 PLC in a lab environment.
- Modeled the system, developed control logic and HMI, and tuned PID parameters so as to achieve stable liquid level control.
- Developed my expertise in industrial automation, PLC programming, and control system tuning.

Simulation program based on M/M/N queuing model Dec. 2024

- Designed and implemented an object-oriented simulation system in Python to model customer flow and service efficiency, utilizing Poisson-based event generation for realistic queue dynamics.
- Built a GUI for parameter control and real-time visualization, enhancing capabilities in simulation modeling and data visualization while delivering actionable operational insights.

Handwriting Numerals Recognition Oct. 2023

- Developed a multi-layer perceptron (MLP) from scratch using NumPy, implementing forward/backward propagation and gradient descent to classify MNIST handwritten digits.
- Achieved 99% test accuracy, validating a robust understanding of neural network fundamentals and efficient numerical computation.

Traffic Management Software Simulation System Mar. 2023 – Apr. 2023

- Developed a graphical user interface by using DOS graphics library and custom mouse interrupt handling with a message loop architecture.
- Designed multi-level menus, dynamic buttons, and data entry forms for interactive user operations.
- Implemented data management with structured file I/O to ensure data consistency and system reliability.

HONORS & AWARDS

Outstanding Completion, National Undergraduate Innovation and Entrepreneurship Training Program 2025
National-level Undergraduate Research Project: Multimodal LLM Agent with Tool Use

Outstanding Student Scholarship, Huazhong University of Science and Technology 2023

SKILLS

Languages : Python, Matlab, C, R, SQL

Tools : Git, Conda, Pytorch, Jupyter Lab, Linux, Simulink, LaTex, HuggingFace, STM32 Platform, Code Computer Studio, Ghidra, IDA Pro, GDB